

AI vs. human: A large-scale analysis of AI-generated fake reviews, human-generated fake reviews and authentic reviews

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ABSTRACT

The rise of fake reviews generated by artificial intelligence (AI) poses significant challenges for online platforms, retailers, and consumers, as advanced generative models like ChatGPT produce fake reviews. More seriously, unlike human-generated fake reviews, which exhibit psychological cues, AI-generated fake reviews lack deceptive intent and instead reflect algorithmic patterns that are increasingly difficult to detect. This study investigates the linguistic distinctions among AI-generated fake reviews, human-generated fake reviews, and authentic reviews by integrating the theoretical model of textual differences and AI characteristics. Analyzing 714,016 reviews through text analysis, entropy weight method and hypothesis testing, this research finds that AI-generated fake reviews demonstrate higher comprehensibility, lower levels of specificity, exaggeration, and negligence compared to human-generated fakes and authentic reviews. These findings challenge the existing paradigms in fake review research. This means that in traditional detection, the situations where there are AI-generated fake reviews that do not conform to rules should be given further consideration. Additionally, AI-generated fake reviews exhibit significantly higher mechanicalness and lower empathy than authentic reviews, providing actionable linguistic cues for identification. These findings highlight the need for revised detection methods that account for AI characteristics. Theoretically, it innovatively combines the textual differences model with AI capabilities to investigate AI-generated fake reviews, enriching the fake review detection literature. Practically, it offers e-commerce platforms and consumers enhanced detection metrics to improve algorithmic accuracy in identifying AI-generated deception.

1. Introduction

The emergence of artificial intelligence (AI) generation technology has introduced a new obstacle to online ecosystems: the proliferation of indistinguishable AI-generated fake reviews (Amos and Zhang, 2024; Palmer, 2023). Notably, Amazon has reported a surge of ChatGPT-generated fake reviews that are overwhelming product review sections (Gault, 2023; Palmer, 2023). Due to the advanced generative and mimetic capabilities of generative AI, consumers are faced with perplexing and challenging-to-identify AI-generated fake reviews (CHEQ, 2021; Salminen et al., 2022). In this context, AI-generated fake reviews erode user confidence in online reviews, and their uncontrollable proliferation negatively impacts seller performance and the healthy online business environment (Shukla and Goh, 2024). Given these challenges, determining the linguistic characteristics of AI-generated fake reviews for effective identification has emerged as a

pressing research problem.

Although existing studies have devised identification rules or methods based on linguistic cues of fake reviews, the AI-generated fake reviews have posed new challenges, as contemporary methods demonstrate limited efficacy in discerning AI-generated fake reviews (Gault, 2023; Shukla and Goh, 2024). Previous human-generated fake reviews were identified through authors' psychological states and corresponding linguistic cues (Luo et al., 2023; Wu et al., 2020a). Nevertheless, it is important to note that AI may not have deception perceptions and guilt similar to humans when generating fake reviews (Loureiro et al., 2021). Because AI acts by the manipulation of the user, the generation of harmful, and fake information is merely the fulfillment of the user's task (Paul et al., 2023). Meanwhile, generative AI possesses remarkable learning and imitating capabilities, having linguistic abilities that reach human levels (Kim et al., 2023a). Even though existing research has focused on the linguistic cues of fake reviews (Luo et al., 2023; Wu et al.,

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2020a), it has limited understanding of the linguistic differences between AI-generated fake reviews and human-generated fake reviews. Thus, these factors compel us to examine the distinctions between AI-generated fake reviews and human-generated fake reviews. Gaining insight into these differences can clarify the new constraints and issues faced by current detection methods, thereby driving the evolution of the methods.

Meanwhile, the differentiation of AI-generated fake reviews from authentic reviews constitutes a core problem. Research is increasingly centered on distinguishing AI-generated fake reviews by comparing the linguistic differences between authentic reviews and AI-generated fake reviews. [Xylogiannopoulos et al. \(2024\)](#) found that AI-generated fake reviews contained more repetitive and similar textual content when compared to authentic reviews. Additionally, vocabulary and emotional cues can be employed to detect AI-generated fake reviews ([Bader et al., 2023](#)). However, previous studies have failed to incorporate AI characteristics for discerning linguistic differences between authentic reviews and AI-generated fake reviews. The AI characteristics are associated with their interaction behavior ([Pelau et al., 2021](#)) and generated content ([Zhang et al., 2024](#)). For instance, the impartiality and accuracy of AIs contribute to the perception that the generated content possesses higher levels of objectivity ([Henestrosa et al., 2023](#)). Furthermore, the limiting features of generative AI also aid in understanding the AI-generated content (AIGC). For example, biases in the large language models on which generative AI relies can lead to discriminatory and biased language in AIGC ([Fang et al., 2024](#)). Thus, it is essential to explore the differences between AI-generated and authentic reviews based on AI characteristics.

Therefore, we sought to investigate the following research questions: (1) How do the linguistic characteristics of AI-generated fake reviews differ from those of human-generated fake reviews? (2) How to identify AI-generated fake reviews through linguistic cues based on AI characteristics?

The theoretical model of textual differences on online reviews provides a foundational reference for addressing the research questions ([Banerjee and Chua, 2017b](#)). This theoretical model is comprehensive and contains four constructs (comprehensibility, specificity, exaggeration, and negligence), involving 13 indicators. This model is used to examine the linguistic contrast between authentic and fake reviews. This model offers comprehensive insights into distinguishing AI-generated fake reviews from human-generated ones, validating the efficacy of traditional rules. Furthermore, unlike humans who compose reviews based on personal experiences, psychological states, and individual personalities ([Luo et al., 2023](#); [Wu et al., 2020a](#)), generative AI constructs textual content through pre-programmed algorithms and imitation of human expressions ([Jackson and Panteli, 2023](#)). To address this distinction, this study also incorporates AI-specific characteristics as discriminative criteria. Prior research has conceptualized AI capabilities as demonstrating superiority in mechanical tasks while lacking emotional or empathic capacities ([Huang et al., 2019](#); [Huang and Rust, 2018](#)). These capabilities manifest in natural language processing tasks, which can be operationalized through linguistic cues and textual features ([Banh and Strobel, 2023](#)).

Guided by theoretical foundations, this study conducts a large-scale empirical investigation based on 714,016 reviews, employing text analysis and the entropy weight method to quantify sub-dimensional linguistic cues and variables. Through empirical comparisons, we systematically analyze linguistic differences among AI-generated fake reviews, human-generated fake reviews, and authentic reviews. The findings reveal that AI-generated fake reviews exhibit higher comprehensibility, lower exaggeration, and lower negligence compared to both human-generated fake reviews and authentic reviews, which sharply contrast with conventional understandings of fake review characteristics. This suggests that existing detection rules may be inadequate for identifying AI-generated fake reviews. Furthermore, when compared to authentic reviews, comparative analysis reveals that AI-generated fake

reviews exhibit significantly higher mechanicalness and lower empathy levels than authentic reviews, providing actionable linguistic cues for detection. These findings not only challenge traditional paradigms in fake review research but also highlight the urgent need to develop new detection factors tailored to AI-generated fake reviews.

The anticipated outcome of this study is to make contributions to both academic research and practical applications. Theoretically, it innovatively integrates the textual differences model with AI capability to examine the distinctive characteristics of AI-generated fake reviews. Existing research has primarily focused on distinguishing AI-generated fake reviews by comparing linguistic differences, failing to account for the distinctions between AI-generated and human-generated fake reviews. Our research not only expands the application boundaries of the theoretical model but also enriches the literature on fake review detection. Furthermore, this study conducts linguistic analysis of AI-generated fake reviews based on AI capabilities, thereby expanding their interpretive scope. By incorporating AI's functional abilities and developing corresponding textual variables, our research not only provides novel perspectives for fake review detection but also establishes a theoretical foundation for future studies on AIGC. In practice, this study offers actionable insights and methodological recommendations for developers of fake review detection systems. Moreover, the results provide practical guidance for e-commerce platforms, retailers, and consumers in identifying suspicious AI-generated content through these diagnostic linguistic markers.

The remainder of this paper is organized as follows. In Section 2, we provide an overview of the theoretical background. In Section 3, we outline the hypotheses. Section 4 introduces the research design. Section 5 and Section 6 present Study 1 and Study 2 respectively. Section 7 presents the discussions. In Section 8, we elaborate theoretical contributions, practical implications, limitations, and future work.

2. Theoretical background and literature review

2.1. The linguistic characteristics of AIGC

Concerned about the impact of AIGC ([Baek and Kim, 2023](#)), some studies have examined the attributes of AIGC with the aim of extracting insights into its characteristics and fundamental principles ([Elkhatat et al., 2023](#)). AIGC is commonly perceived as possessing greater objectivity and rigor in comparison to human-generated content, albeit at the expense of creativity and readability ([Graefe et al., 2018](#)). [Wu et al. \(2020b\)](#) found that AIGC was similar in quality to human-generated content, but there were still gaps in the emotional and creative aspects. Furthermore, it can be found that AI-generated content may exhibit repetitive or formulaic words or patterns, while human-generated texts are more inclined to be original and innovative ([Mindner et al., 2023](#)). Nevertheless, [Wu and Zheng \(2025\)](#) suggested that generative AI surpasses human performance in managing responses to collaborative and prosocial inquiries.

Furthermore, [Herbold et al. \(2023\)](#) conducted a large-scale study that indicated the quality of the AI-generated essays was significantly better than that written by humans. However, AI-generated essays exhibit a consistent structure, whereas human-generated essays are more flexible. [Zhang et al. \(2024\)](#) suggested that the text quality of AI-generated tourism explanatory content was inferior to that of professional explanatory content.

2.2. The linguistic cues of fake reviews

The linguistic cues of fake reviews have been the key to identifying them, which has prompted numerous scholars to study them from various perspectives by comparing them with authentic reviews ([Wu et al., 2020a](#)). As summarized in [Table 1](#), linguistic cues gradually evolved from word and sentence analysis to textual analysis. Initially, word and sentence analysis easily reveals differences between authentic

Table 1

The summary of literature regarding linguistic cues of fake reviews.

Linguistic cues	Findings	Author(s) (Year)
Simple and short words	Fake reviews prefer use simple and short words	Jindal and Liu (2008)
Length and depth	Fake reviews may be shorter than authentic reviews; Fake reviews are associated with less in-depth and specific analysis	Chen et al. (2021); Birim et al. (2022); Kim et al. (2023b)
Lexicality	Fake reviews contain more verbs, adverbs, pronouns, and pre-determiners	Ott et al. (2011)
Grammar and spelling error	Fake reviews contain less grammar and spelling error	Banerjee and Chua (2021)
Repeat content	Fake reviews contain more repeat content	Jindal and Liu (2008); Plotkina et al. (2020)
Emotional terms and cues	Fake reviews are more inclined to use emotional terms	Ott et al. (2013); Banerjee and Chua (2017a); Wang et al. (2022)
Affective cues	Fake reviews contain more affective cues.	Banerjee and Chua (2014); Li et al. (2020)
Extreme cues	Fake reviews tend to be more extreme	Luca and Zervas (2016)
Social cues	Fake reviews contain more social cues	Van Swol and Braun (2014); Li et al. (2020)
Comprehensible cues	Fake reviews are less comprehensible	Banerjee and Chua (2017a); Moon et al. (2021)
Specific cues	Fake reviews are less specific	Banerjee and Chua (2017a)
Negligent cues	Fake reviews are linguistically negligent	Banerjee and Chua (2017a)
Readability	There are two opposing views	Luo et al. (2023)
Linguistic conflict	Fake reviews contain more linguistic conflicts	Shan et al. (2021)
Repetition	AI-generated fake reviews are more repetitive	Xylogiannopoulos et al. (2024)

and fake reviews, such as length, depth, lexicality, and errors. Because fake reviewers lack experience and to save time, they use simple words and generate succinct and in-depth fake reviews (Birim et al., 2022; Chen et al., 2021; Jindal and Liu, 2008; Plotkina et al., 2020). Verbs, pronouns, and adverbs in fake reviews are often representative of the lack of actual experience and imagination of the fake reviewer (Ott et al., 2011). In addition to the above, fake reviews tend to contain fewer grammatical and spelling errors (Banerjee and Chua, 2021).

In addition, previous research found that the cues of fake reviews are often inextricably linked to the psychological state of reviewers. Scholars acknowledged that compared to authentic reviews, fake reviews tend to exhibit a higher degree of exaggeration and employ a greater number of emotionally charged phrases (Luca and Zervas, 2016; Ott et al., 2013; Wang et al., 2022). This also means that more clues indicating exaggeration and emotion can be found in the fake reviews (Banerjee and Chua, 2014; Li et al., 2020). The degree of exaggeration also varies depending on the valence of the reviews (Martinez-Torres and Toral, 2019). Furthermore, Banerjee and Chua (2017a) compiled a model of linguistic features that argues for less comprehensibility, less specificity, higher levels of exaggeration, and less linguistic negligence in fake reviews. Li et al. (2020) also proposed a framework of psychological cues based on interpersonal deception theory, indicating that fake reviews have fewer perceptual cues and more emotional and social cues. In addition to the above contents, readability and contradiction have also been recognized as important indicators for detecting fake reviews. There has been an ongoing discussion about readability, with different studies giving opposing viewpoints (Luo et al., 2023). Shan et al. (2021) posited that users are likely to post reviews with inconsistent or even conflicting content for reasons such as the fake experience, proposing 22 linguistic features that test for comment conflict for the identification of fake reviews.

Notably, existing research found differences between authentic reviews and AI-generated fake reviews, with findings suggesting that AI-

generated fake reviews are more repetitive compared to authentic reviews (Xylogiannopoulos et al., 2024). However, existing studies have not yet examined the differences between AI-generated fake reviews and human-generated fake reviews, nor have they considered the characteristics of AI in identifying AI-generated fake reviews.

2.3. The theoretical model of textual differences

To identify linguistic cues to distinguish between authentic and fake reviews, Banerjee and Chua (2017b) built a theoretical model based on the information manipulation theory, the leakage theory, the self-presentational theory, and the reality monitoring theory. This theoretical model contains four constructs: comprehensibility, specificity, exaggeration and negligence. This model is considered effective in analyzing online reviews, especially in discerning authentic reviews and fake reviews, which is shown in Table 2.

Firstly, comprehensibility of a review refers to the extent to which it is easy to understand (Banerjee and Chua, 2017b). Building on the information manipulation theory (McCormack, 1992), authentic and fake information can be distinguished by the information quantity and expression methods. In addition, self-presentation theory emphasizes that individuals who provide fake information may enhance the credibility of their arguments through complex language (DePaulo et al., 2003). In this case, those who write fake reviews may use complex language. In this model, comprehensibility is considered to include readability, word familiarity, and structural features. Texts that are readable, use familiar words, and avoid lengthy words are regarded as comprehensible (Banerjee and Chua, 2017a).

Secondly, specificity of a review refers to the extent of details indicated in reviews (Bond Jr and DePaulo, 2006). Information manipulation theory suggests that authentic information and fake information can also be distinguished based on information quality and relevance (McCormack, 1992). Furthermore, reality monitoring theory posits that the distinction between true and fake information lies in the perception and contextual details they convey (Johnson and Raye, 1981). On this basis, specificity is considered to contain informativeness, perceptual details, and contextual details.

Thirdly, exaggeration of a review refers to the writing style used in

Table 2

The theoretical model of textual differences.

Construct	Definition	Sub-dimensions	Measurement
Comprehensibility	The extent to which reviews are clear and easy to understand	Readability Word familiarity Structural features	Texts that are readable, use familiar words, but avoid lengthy words are regarded as comprehensible.
Specificity	The extent of details indicated in reviews	Informativeness Perceptual details Contextual details	Text is informative and contains more perceptual and contextual details, which is regarded as specificity.
Exaggeration	The ways specific types of words are used to construct sentences in reviews convincingly	Affectiveness Tenses Emphases Syntactic features	Affective with a temporal focus on present or future, containing several punctuations but few function words, are deemed exaggerated.
Negligence	Cues that leak out due to the cognitive challenges involved in writing fictitious reviews	Self-references Uncertainty Cognitive cues	Text with few self-references but rich in uncertainty and cognitive words are deemed to reflect negligence.

the entry to sound convincing. According to the leakage theory, although both authentic reviewers and fake reviewers attempt to persuade others, the latter may inadvertently exaggerate their viewpoints (Ekman and Friesen, 1969). Furthermore, the self-presentation theory suggests that individuals writing fake reviews may rely on exaggerated rhetoric in an effort to enhance their credibility. Exaggeration is considered to be related to affectiveness, tenses, emphases, and syntactic features. In general, affective, hyperbolic reviews with a temporal focus on present or future, containing several punctuations but few function words, are deemed exaggerated (Banerjee et al., 2017; Tausczik and Pennebaker, 2010).

Fourthly, negligence of reviews refers to clues for deception detection that are coincidentally leaked out in the entry. Generally speaking, based on the leakage theory, writing fake reviews is more challenging than writing true reviews (Ghose and Ipeirotis, 2010). Fake reviewers may be negligent when carrying out this task. In addition, according to the reality monitoring theory, real information and fictional information are distinguished based on the extent of psychological processing engaged by the information provider. Based on this, negligence is related to three sub-dimensions: self-references, uncertainty words, and cognitive words. Reviews that have few self-references but are rich in uncertainty and cognitive words are deemed to reflect negligence.

2.4. The AI capabilities

The utilization of AI has witnessed a notable rise in its application (Yu et al., 2024). To explore more fully and rigorously the AI characteristics, Huang et al. (2019) proposed that AI has rapidly evolved from mechanical capabilities but is still lacking in feeling capabilities. On the one hand, mechanical capability involves the capability of AIs to automate routine repetitive tasks, which they can do based on predetermined rules and past knowledge (Huang and Rust, 2018). For example, hotel service robots can provide outputs in a fixed paradigm for specific problems (Xu et al., 2023). Naturally, the evolution of AI extends beyond its capability to perform monotonous and repetitive activities, but maintaining its mechanical capabilities contributes to the accuracy and dependability of the outcomes produced (Huang et al., 2019). On the other hand, feeling embodies the capability to recognize, imitate, and respond appropriately to human emotions. Specifically, the AI possesses the capability to comprehend human behavior and offer responses with a sense of empathy. Although AI is now striving to achieve a perceptual capability similar to that of humans, AI has yet to bridge the substantial disparity in perceptual capabilities between machines and humans (Huang et al., 2019).

The content generated by the AI can reflect the underlying logic and strategy (Campbell et al., 2022). The superiority and limitations of AI in different aspects can influence their generated content. This study aims to essentially differentiate AI-generated fake reviews from authentic reviews generated by human beings through AI characteristics.

3. Hypotheses development

Building on the theoretical background and literature review, this study compares AI-generated fake reviews, human-generated fake reviews and authentic reviews through linguistic cues. Firstly, the theoretical model contains four constructs (comprehensibility, specificity, exaggeration and negligence) (Banerjee and Chua, 2017b), which is previously used to distinguish between human-generated fake reviews and authentic reviews. By applying this model, the present study examines linguistic differences among AI-generated fake reviews, human-generated fake reviews, and authentic reviews, aiming to determine whether traditional detection rules remain applicable to AI-generated content. Furthermore, this study focuses on the AI capabilities to distinguish between AI-generated fake reviews and authentic reviews. To enhance detection precision, we propose novel linguistic cues based on AI's strengths and limitations as key discriminative

features.

3.1. Comprehensibility

Comprehensibility is defined as the degree to which a review is easily understood, including readability, word familiarity, and structural features (Banerjee and Chua, 2017b). Typically, comprehensibility is predicated upon the intricacy of sentence structures and the selection of lexicon. The utilization of infrequent vocabulary and intricate sentence structures may reduce comprehensibility (Liang et al., 2019). Human-generated fake reviews are often short (Banerjee and Chua, 2014), and are considered easy to understand. Many deceivers need to complete a large number of fake review generation tasks, and they save time by completing reviews in shorter word counts (Luo et al., 2023). Furthermore, it has been observed that several sellers incentivize customers to fabricate counterfeit evaluations by offering monetary incentives, which customers may hastily submit without much thought (Chen et al., 2021). Nevertheless, AI has shown superior performance in content generation and generates content in a comprehensive manner (Herbold et al., 2023). In this context, it is imperative that AI should be provided to supplement of the copious and superior quality content. Therefore, we hypothesize that.

H1(a). AI-generated fake reviews are more comprehensible than human-generated fake reviews.

In addition, AI-generated fake reviews, a type of AI-generated content, tend to provide rigorous and structured content that is considered less comprehensible and readable (Graefe et al., 2018). Similarly, Markowitz et al. (2024) discovered that AI-generated text had a more analytical style and was more descriptive and less readable than human-generated text. However, authentic reviews are often considered simple to read (Wang and Kuan, 2022). Therefore, we hypothesize that.

H1(b). AI-generated fake reviews are less comprehensible than authentic reviews.

3.2. Specificity

Specificity refers to the richness in details of a review, including informativeness, perceptual details, and contextual details. Numerous prior research endeavors have established that the specificity of reviews is a significant indicator for detecting fake reviews (Moon et al., 2021). AI possesses a massive database to support content generation, and its powerful collection and integration capabilities give it a strong compilation capability (Lund et al., 2023). AI will compose or even duplicate rich content on demand, which aids its detail in writing fake reviews (Fosso Wamba et al., 2024). However, human-generated fake reviews are much less detailed. This is because fake reviewers are incapable of providing an in-depth description of the experience object and process, particularly with respect to its informativeness and perception (Banerjee and Chua, 2021). On the one hand, one of the challenges associated with fake reviews is the limited ability to communicate comprehensive and valuable information (Banerjee and Chua, 2014). Fake reviewers may involve imitating pre-existing reviews or providing a general overview of the experience rather than a comprehensive analysis. On the other hand, counterfeit reviewers are incapable of delineating the intricacies of their encounter due to their absence of genuine experience (Li et al., 2020). Therefore, we hypothesize that.

H2(a). AI-generated fake reviews are more specific than human-generated fake reviews.

However, authentic reviews are typically grounded in consumers' personal experiences, enabling them to provide detailed descriptions of product functionalities and experiences (Kakaria et al., 2024). Specifically, these reviews often include personalized details, such as specific usage scenarios and emotional responses to product performance (Li

et al., 2022). These unique elements arise from users' individual perceptions and lived experiences. In contrast, AI-generated reviews are constructed through pattern recognition from large-scale text corpora, lacking the cognitive foundation of real-world encounters (Wani et al., 2024). While generative AI can mimic surface-level linguistic patterns, it frequently produce overly generic phrasing rather than context-specific details (Lai and Nissim, 2024). Therefore, we hypothesize that.

H2(b). AI-generated fake reviews are less specific than authentic reviews.

3.3. Exaggeration

Exaggeration refers to being perceived with a convincing linguistic style, usually associated with sentiments (Banerjee and Chua, 2017b). It contains four indicators: affectiveness, tense, emphases, and syntactic features (Banerjee et al., 2017). The attribute of exaggeration is readily apparent to readers when perusing a review, as it allows for the assessment of the emotional inclination and intensity of the text (Shan et al., 2021; Yang et al., 2021). There have been many studies that confirm that fake reviews contain more emotional linguistic cues or more intense emotional cues (Li et al., 2020; Wang et al., 2022). According to Ott et al.'s research, reviews that exhibit intense emotions are highly indicative of being fake reviews (Ott et al., 2013). In addition, certain researchers have posited that fake reviews have a tendency to exhibit greater levels of exaggeration when compared to authentic reviews (Banerjee and Chua, 2017b). On the one hand, individuals who engage in fake reviewing endeavor to persuade other users through the use of hyperbolic language. On the other hand, the utilization of hyperbolic language may compensate for the lack of observation and specificity (Banerjee, 2022). Therefore, we hypothesize that.

H3(a). AI-generated fake reviews are less exaggerated than human-generated fake reviews.

Although authentic reviews are considered less exaggerated than fake reviews (Banerjee, 2022), some users may also use slightly exaggerated expressions when providing genuine reviews (Kapoor et al., 2021). However, it can be observed that the capacity of AI to convey emotions is comparatively inferior to that of human beings (Bakpayev et al., 2022; Han et al., 2022). In this context, it is difficult for generative AI to use exaggerated tones during the generation process. Therefore, we hypothesize that.

H3(b). AI-generated fake reviews are less exaggerated than authentic reviews.

3.4. Negligence

Negligence means the extent to which a review inadvertently leaks clues to deception detection, including self-references, uncertainty words and cognitive words. Deception is often related to psychological states and can reveal clues of negligence. It has been argued that fake reviews reveal more negligent content because they may reveal the reviewer's inner guilt and uncertainty (Banerjee et al., 2017). However, different from previous frauds that involve psychological attribution, generative AI is considered not to contain similar beliefs or psychological states (Tarsney, 2025). In other words, AI may not have deception perceptions and guilt similar to humans when generating fake reviews (Loureiro et al., 2021). In this case, it would not realize that it is deceiving others and thus revealing itself.

H4(a). AI-generated fake reviews show less negligence than human-generated fake reviews.

Meanwhile, generative AI is confronted with the issue of knowledge hallucination. (Cao et al., 2025). Although AI may provide incorrect or fake information, it presents logical arguments that make it seem correct, thereby confusing the reader (Van Dis et al., 2023). Furthermore,

authentic reviews are freely provided by users in an online environment, and they may contain users' personal biases or linguistic errors (Filieri, 2016). Nevertheless, AI-generated reviews will be more rigorous and avoid obvious issues. In such cases, AI may not easily reveal clues of its inaccuracies. Therefore, we hypothesize that.

H4(b). AI-generated fake reviews show less negligence than authentic reviews.

3.5. Mechanicalness

Mechanicalness was employed to describe the connection between the brain and computers, recognized as mechanisms and sensations associated with formality and logical sense. Formality implies complete computation and processing, which entails transforming entities from initial conditions (inputs) to termination conditions (outputs). Logical sense signifies following specific rules or symbols to accomplish calculations or information processing (Shagrir, 2006). In terms of expression, mechanicalness is typically conveyed through consistent and coherent elements (Adorno and Paddison, 1982).

For AI generation tools, it is worth noting that while the content to be generated may exhibit significant diversity, the method of production often adheres to a specific paradigm. Content generation is typically dependent on program parameters and is accompanied by specific answer generation algorithms and predetermined answers (Jackson and Panteli, 2023). In addition, to ensure the accuracy and standardization of the generated content, AI-generated tools may adhere to certain static rules. These pre-programmed and static rules will be represented in the AI-generated content, rendering AI-generated content lacking freshness and innovation, yet structured and logical (Mindner et al., 2023). These characteristics contribute to enhancing the mechanicalness of AI-generated content. For many projects, the mechanical representation of AI-generated content makes it more readable and easy to understand (Liu et al., 2023).

However, online reviews, as individual-to-individual communication in an online environment, are distinguished from formal textual content in terms of both content and expression form (Zhang and Niu, 2024). As a result, online reviews are characterized by a greater degree of flexibility and informality in their expression. During reviewing, honest reviewers prioritize presenting a comprehensive narrative rather than employing persuasive techniques to convince others of flawlessness (Ansari and Gupta, 2021). Therefore, we believe that AI-generated fake reviews are superior mechanically and differ significantly from authentic reviews. Hence, this work proposes the following hypothesis.

H5. AI-generated fake reviews are more mechanical than authentic reviews.

3.6. Empathy

Empathy can be defined as a combination of emotional reactions and the cognitive understanding of other people's experiences and feelings (Simon, 2013). Research shows that online reviews can stimulate potential customers to develop empathy towards the reviews. Every emotion they experience while reading the reviews will influence their feelings or behaviors (Hossain and Rahman, 2022). Currently, AI is able to demonstrate the ability to feel similarly to humans to convey an understanding of users (Mari et al., 2024). For example, generating AI-generated content can mimic the perception of humans to illustrate that it contains feelings of pleasure or surprise (Hitsuwari et al., 2023).

Despite the improving ability of AI to express emotions, it is still widely perceived as a cold and emotionless entity because it cannot fully replicate human empathy (Perry, 2023). Compared with human-generated content, AI-generated content lacks more connection and empathy with users (Zhang et al., 2024). Furthermore, authentic reviews can encompass complex or even contradictory emotions because the reviewer may be satisfied with certain elements (Banerjee

and Chua, 2017b). For AI, these abundant and complex emotions are difficult for them to understand and express. Hence, this work proposes the following hypothesis.

H6. AI-generated fake reviews are less empathetic than authentic reviews.

4. Research design

4.1. Overall research design

This study employs a two-phase experimental approach, as illustrated in Fig. 1. Initially, to investigate the discriminative performance of linguistic cues proposed in prior literature for identifying AI-generated fake reviews, Study 1 systematically compares AI-generated fake reviews, human-generated fake reviews, and authentic reviews within the theoretical model. Specifically, we quantify linguistic features across 13 sub-dimensions using text analysis algorithms and apply entropy weighting to evaluate four key variables. Differences among the three review types are then assessed via one-way ANOVA. In addition, in Study 2, taking into account the influence of AI capabilities on its generation paradigms, this study proposes two variables to distinguish between AI-generated fake reviews and authentic reviews. Using text analysis techniques and independent samples t-tests, this study explores the disparities between AI-generated fake reviews and authentic reviews along two key dimensions: mechanicalness and empathy.

4.2. Data collection

The authentic reviews and human-generated fake reviews for this

study came from the YelpZIP dataset provided by Yelp website (Yelp, 2019), and the AI-generated fake reviews were collected from ChatGPT 4.0.

The human-generated fake reviews and authentic reviews were sourced from the YelpZIP dataset. This dataset primarily collects restaurant reviews from multiple U.S. states with consecutive ZIP codes on the Yelp website, making it the largest fake review dataset (Rayana and Akoglu, 2015). The Yelp website (www.Yelp.com) is an internet-based review platform that encompasses evaluations provided by consumers pertaining to various services, such as restaurants and bars, and receives approximately 130 million unique visitors per month (Luca and Zervas, 2016; Yelp, 2021). Its significant influence has led it to be frequently affected by fake reviews. As a result, Yelp built algorithms to identify fake reviews, which has been claimed to be highly accurate (Kumar et al., 2022). On this website, fake reviews identified by algorithms are marked. This dataset collects restaurant reviews from 2004 to 2015, including authentic reviews and labeled fake reviews, which avoids the inclusion of AI-generated fake reviews. The dataset is widely used for fake review detection and online review research (Li et al., 2023; Pan et al., 2024).

In addition, to collect AI-generated fake reviews, we employed a structured data collection approach using Python code to interface with ChatGPT 4.0 through API (Kocon et al., 2023). This method was chosen for its alignment with real-world scenarios, as validated through three aspects to ensure data reliability. First, the selection of ChatGPT 4.0 to collect AI-generated fake reviews is realistically representative. ChatGPT has been reported as the primary tool behind AI-generated fake reviews on digital platforms (Gault, 2023; Palmer, 2023). This widespread usage reinforces the credibility of our dataset, making it highly representative of real-world fake reviews encountered in online

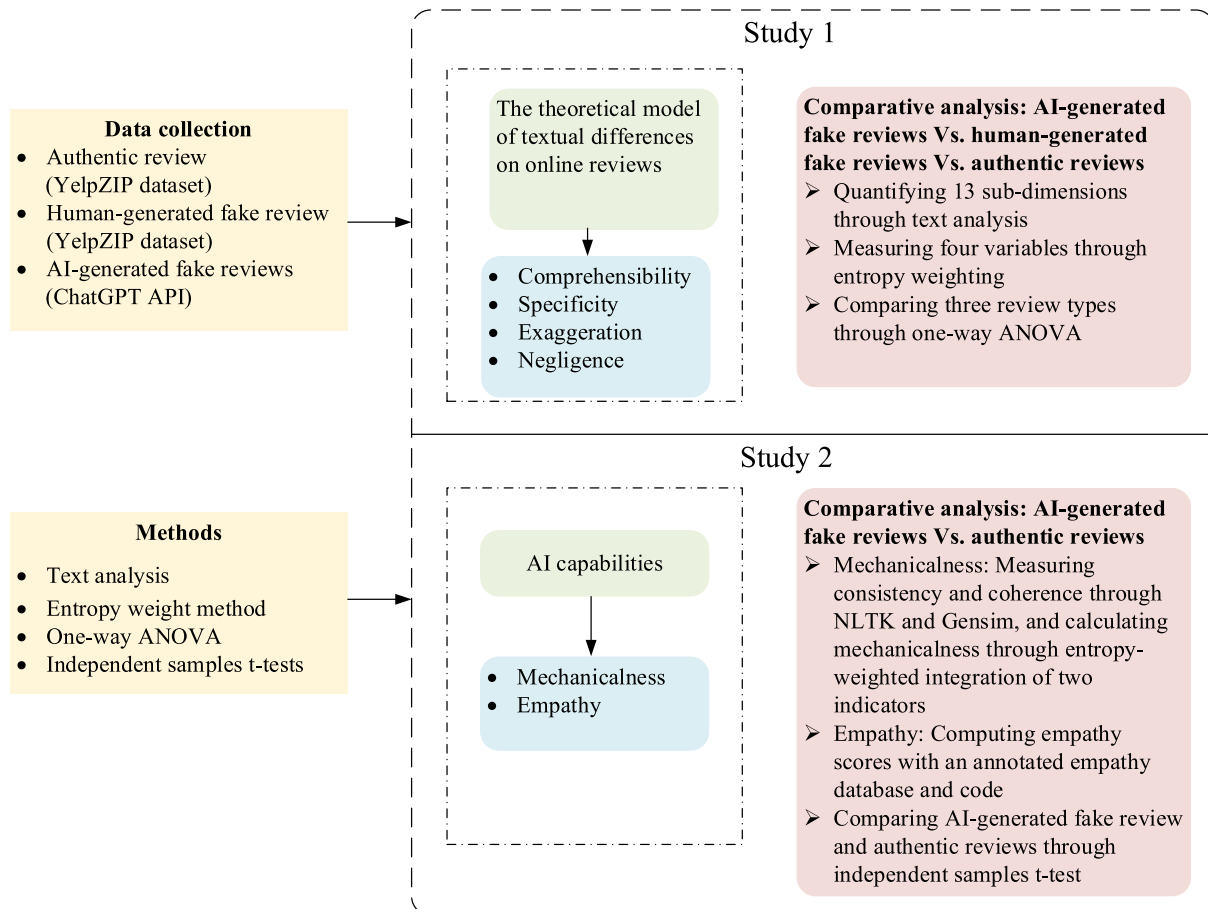


Fig. 1. Overall research design.

environments.

Second, existing academic research on AI-generated fake reviews generally employs ChatGPT 4.0 to collect AI-generated fake reviews via API (Amos and Zhang, 2024; Xylogiannopoulos et al., 2024). As OpenAI's officially interface for external integration, the API enables systematic large-scale data generation while maintaining consistent output quality (Paredes et al., 2023). This automated approach minimizes human intervention and reduces sampling bias. Therefore, this study follows established research methodologies by implementing Python scripts to interact with ChatGPT's API for generating AI-generated fake reviews (Kocon et al., 2023). It is worth noting that to address potential biases and ensure output diversity, we utilized over 70 unique API keys across multiple OpenAI accounts. This approach mitigates potential bias from any single API instance and facilitates large-scale AI generation data collection.

Third, to better align the AI-generated fake reviews with real-world scenarios, contextual prompts for data collection were meticulously designed using real-world restaurant data from the YelpZIP dataset, which encompasses authentic restaurant names and geospatial coordinates (Rayana and Akoglu, 2015). This methodology enables AI-generated fake reviews to accurately reflect real-world dining experiences.

Totally, after removing duplicate data and outliers, we aggregated 714,016 reviews into three groups, with 117,859 reviews in the AI-generated fake review group, 519,069 reviews in the authentic review group and 77,088 reviews in the human-generated fake review group. Then, we extracted text features and quantify them according to the research hypothesis to base the hypothesis testing.

5. Study 1

To systematically explore the nuanced linguistic differences among AI-generated fake reviews, human-generated fake reviews, and authentic reviews in Study 1, we strategically selected one-way ANOVA as our analytical method as it allows simultaneous comparisons of means across three or more independent groups. This method has been adopted in linguistic analysis (Kirilenko et al., 2024), particularly in investigations examining the contrast between fake and authentic reviews (Banerjee, 2022; Hlee et al., 2021). Existing research has repeatedly demonstrated that one-way ANOVA excels at pinpointing significant differences in key language features such as sentiment polarity and word length when contrasting fake and authentic reviews (Kim, 2024; Plotkina et al., 2020).

Specifically, to compare the three distinct review groups, one-way ANOVA's post-hoc test (Tukey test) was employed which has been verified to offer a statistically robust approach to conduct pairwise comparisons of linguistic features in prior research (Plonsky et al., 2015). This method not only enables to statistically confirm the presence of significant differences among three types of reviews but also to conduct detailed pairwise comparisons. By leveraging this method, we can also accurately identify which specific linguistic sub-dimensions contribute to these disparities. The approach is particularly well-suited to our research context as it enables us to systematically analyze the differences in linguistic features across review categories. The application of one-way ANOVA with post-hoc test (Tukey test) provides a solid analytical foundation, allowing us to make well-founded inferences about the three review groups.

5.1. Measurement of variables

To better serve the experimental needs, this study utilizes text analysis algorithms and text analysis tools to measure 13 sub-dimensions (Banerjee and Chua, 2017a). The measurement methods are displayed in Table 3. In order to calculate the constructs for the combined sub-dimensions, the entropy weight method is adopted to conduct combined calculations for the four constructs of the

Table 3

The measurement of sub-dimensions.

Construct	Sub-dimensions	Measurement
Comprehensibility	Readability	Using indicators such as Automated-Readability Index, Coleman-Liau Index, Flesch-Kincaid Grade Level, Gunning-Fog Index, and Rate Index, The average score of the indicators.
	Word familiarity	Compared against the Dale-Chall lexicon of familiar words to measure the proportion of easily-recognizable words.
Specificity	Structural features	The reciprocal of the proportion of words with more than 10 letters
	Informativeness	The proportion of the following eight POS tags: nouns, adjectives, prepositions, articles.
	Perceptual details	The proportion of words representing perception was measured utilizing Linguistic Inquiry and Word Count 2022 (LIWC2022) (Ryan L. Boyd, 2022).
	Contextual details	The proportion of words representing perception was measured utilizing LIWC2022, including spatial (e.g. down), and temporal (e.g. until) words
Exaggeration	Affectiveness	The proportion of words representing positive emotion (e.g. nice) and negative emotion (e.g. ugly) words utilizing LIWC2022.
	Tenses	The proportion of past, present, and future tense words was measured using LIWC2022.
	Emphases	The proportion of capitalized words and proper terms.
	Syntactic features	The proportion of these punctuation marks: ellipses, emoticons, exclamation marks, question marks, and all punctuation used.
Negligence	Self-references	The reciprocal of the proportion of first-person singular (e.g., I) and plural (e.g., we) which was measured using LIWC2022.
	Uncertainty	The proportion of modal verbs, filler words, and tentative words.
	Cognitive cues	The proportion of cognition words was measured using LIWC2022.

sub-dimensions. After standardizing the data of the sub-dimensions through the entropy weight method, weights are assigned to them (Chen et al., 2024). Since the sub-dimension measurement data obtained from text analysis are continuous data rather than scale data, and each dimension has different weights, they do not conform to the assumption of Tau-equivalence. Therefore, this study uses McDonald's omega to test the internal consistency of the sub-dimensions. This index is widely recognized as a more reasonable and robust index for handling multi-dimensional scales and mixed-format data (Flora, 2020; Hayes and Coutts, 2020; Watkins, 2017), and it is not restricted by the Tau-equivalence assumption (Dunn et al., 2014). This indicator has been used to test the reliability of data distinguishing fake information (Sindermann et al., 2021). In this research, the McDonald's omega values for these four constructs are 0.72, 0.74, 0.72, and 0.71 in sequence, all of which meet the criteria for McDonald's omega, which indicates adequate reliability and consistency (Romero-Rodríguez et al., 2020).

5.2. Results

In Study 1, we conducted a one-way ANOVA with Post-Hoc test (Tukey test) to examine the linguistic differences among AI-generated fake reviews, human-generated fake reviews and authentic reviews. As shown in Table 4 and Fig. 2, the results indicate that these three types of reviews differ in comprehensibility, specificity, exaggeration, and negligence. The post-hoc analysis suggested that the comprehensibility

Table 4
One-way ANOVA results.

Variable	Review group	Mean	Mean difference	p
Comprehensibility	AI-generated fake reviews	7.26		
	Human-generated fake reviews	5.53	-1.73	0.00***
	Authentic reviews	5.58	-1.68	0.00***
Specificity	AI-generated fake reviews	4.68		
	Human-generated fake reviews	5.62	0.94	0.00***
	Authentic reviews	5.69	1.01	0.00***
Exaggeration	AI-generated fake reviews	7.74		
	Human-generated fake reviews	9.43	1.69	0.00***
	Authentic reviews	9.08	1.34	0.00***
Negligence	AI-generated fake reviews	3.13		
	Human-generated fake reviews	4.59	1.46	0.00***
	Authentic reviews	4.57	1.44	0.00***

Note: ***Significant at .001 level; **Significant at 0.01 level; *Significant at .05 level.

of AI-generated reviews may be greater than that of human-generated fake reviews and authentic reviews ($\text{Mean}_{\text{AI-generated fake reviews}} = 7.26$, $\text{Mean}_{\text{Human-generated fake reviews}} = 5.53$, $\text{Mean}_{\text{authentic reviews}} = 5.58$, $p < 0.01$), so H1(a) was supported. However, it is surprising that the specificity of AI-generated fake reviews is significantly weaker than that of human-generated fake reviews and authentic reviews ($\text{Mean}_{\text{AI-generated fake reviews}} = 4.68$, $\text{Mean}_{\text{Human-generated fake reviews}} = 5.62$, $\text{Mean}_{\text{authentic reviews}} = 5.69$, $p < 0.01$). Thus, H2(b) was supported. Furthermore, as predicted, the exaggeration ($\text{Mean}_{\text{AI-generated fake reviews}} = 7.74$, $\text{Mean}_{\text{Human-generated fake reviews}} = 9.43$, $\text{Mean}_{\text{authentic reviews}} = 9.08$, $p < 0.01$) and negligence ($\text{Mean}_{\text{AI-generated fake reviews}} = 3.13$, $\text{Mean}_{\text{Human-generated fake reviews}} = 4.59$, $\text{Mean}_{\text{authentic reviews}} = 4.57$, $p < 0.01$) are significantly lower for AI-generated reviews than human-generated fake reviews and authentic reviews. Thus, H3(a), H3(b), H4(a) and H4(b) were supported.

To explain in a more fine-grained manner the linguistic differences among AI-generated fake reviews, human-generated fake reviews, and authentic reviews, we further analyzed their linguistic disparities across 13 sub-dimensions. In terms of readability, word familiarity, informativeness, emphases, and self-references, it was found that AI-generated fake reviews outperformed human-generated fake reviews and authentic reviews. However, in terms of other dimensions, AI-generated fake reviews were weaker than human-generated fake reviews and authentic reviews. The results are listed in Table 5.

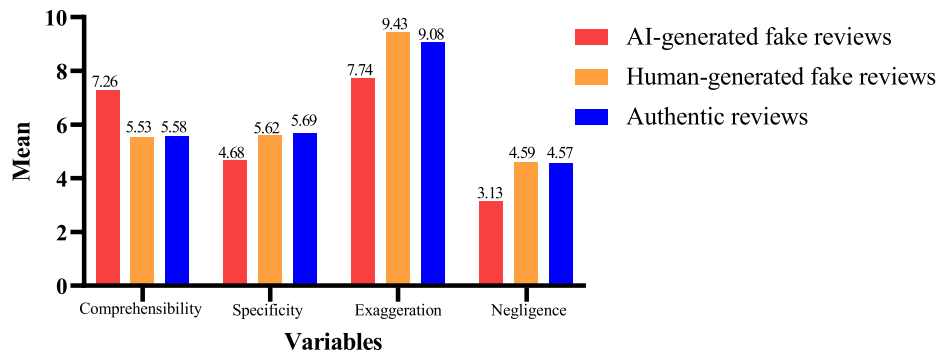


Fig. 2. Linguistic differences among three types of review.

6. Study 2

Study 2 further investigates how AI-generated fake reviews can be distinguished from authentic reviews based on AI characteristics. Based on the dataset collected in Study 1, the reviews were partitioned into two groups: an authentic review group ($N = 519,069$) and an AI-generated fake review group ($N = 117,859$).

6.1. Measurement of mechanicalness and empathy

Mechanicalness is mainly demonstrated through linguistic expressions that are derived from pre-programmed programs and static rules (Jackson and Panteli, 2023; Mindner et al., 2023), which can be measured by textual consistency and coherence (Adorno and Paddison, 1982). Mechanicalness can be obtained by calculating the cohesion score and consistency score (Zaparniuk et al., 1995). This study applies Python's NLTK and the gensim model to separately calculate the consistency and coherence of each review. The mechanicalness is calculated by combining two indicators through the entropy weight method.

Empathy is associated with an important psychological process that facilitates human communication (Zhou et al., 2021). Existing studies have proposed measuring the empathy degree of text through empathy computing. We computed empathy scores by analyzing the collected reviews with an annotated empathy database and code (Sharma et al., 2020).

6.2. Results

In Study 2, we conducted an independent samples *t*-test to examine the linguistic differences between AI-generated fake reviews and authentic reviews, and the results are indicated in Table 6 and Fig. 3. As predicted, the mechanicalness of AI-generated reviews may be greater than that of authentic reviews ($\text{Mean}_{\text{AI-generated fake reviews}} = 3.39$, $\text{Mean}_{\text{authentic reviews}} = 3.19$, $t = 73.16$, $p < 0.01$), so H5 was supported. However, the empathy is significantly lower for AI-generated reviews than authentic reviews ($\text{Mean}_{\text{AI-generated fake reviews}} = 0.36$, $\text{Mean}_{\text{authentic reviews}} = 0.39$, $t = -39.55$, $p < 0.01$). Thus, H6 was supported.

7. Discussion

AI-generated fake reviews have attracted great attention from researchers. To effectively evaluate the validity of existing detection rules and propose actionable identification criteria, this study investigates the linguistic differences among AI-generated fake reviews, human-generated fake reviews, and authentic reviews.

Firstly, according to a series of studies, while AI-generated fake reviews demonstrate unexpectedly high comprehensibility compared to human-generated fakes and authentic reviews, their lower specificity aligns with previous research results. On the one hand, the results indicate that AI-generated fake reviews exhibit higher readability

Table 5
One-way ANOVA results^a.

Sub-dimensions	Review group	Mean	Mean difference	p
Readability	AI-generated fake reviews	10.18		
	Human-generated fake reviews	6.52	-3.66	0.00***
	Authentic reviews	6.65	-3.53	0.00***
Word familiarity	AI-generated fake reviews	9.81		
	Human-generated fake reviews	8.02	-1.79	0.00***
	Authentic reviews	8.06	-1.75	0.00***
Structural features	AI-generated fake reviews	0.96		
	Human-generated fake reviews	0.98	0.02	0.00***
	Authentic reviews	0.98	0.02	0.00***
Informativeness	AI-generated fake reviews	0.62		
	Human-generated fake reviews	0.49	-0.13	0.00***
	Authentic reviews	0.49	-0.13	0.00***
Perceptual details	AI-generated fake reviews	8.95		
	Human-generated fake reviews	9.12	0.17	0.00***
	Authentic reviews	9.51	0.56	0.00***
Contextual details	AI-generated fake reviews	6.88		
	Human-generated fake reviews	10.31	3.43	0.00***
	Authentic reviews	10.15	3.27	0.00***
Affectiveness	AI-generated fake reviews	6.70		
	Human-generated fake reviews	8.59	1.89	0.00***
	Authentic reviews	7.26	0.56	0.00***
Tenses	AI-generated fake reviews	7.91		
	Human-generated fake reviews	10.82	2.91	0.00***
	Authentic reviews	10.88	2.97	0.00***
Emphases	AI-generated fake reviews	8.16		
	Human-generated fake reviews	2.20	-5.96	0.00***
	Authentic reviews	3.09	-5.07	0.00***
Syntactic features	AI-generated fake reviews	11.98		
	Human-generated fake reviews	19.73	7.75	0.00***
	Authentic reviews	19.51	7.53	0.00***
Self-references	AI-generated fake reviews	0.54		
	Human-generated fake reviews	0.37	-0.17	0.00***
	Authentic reviews	0.31	-0.23	0.00***
Uncertainty	AI-generated fake reviews	0.01		
	Human-generated fake reviews	0.02	0.01	0.00***
	Authentic reviews	0.02	0.01	0.00***
Cognitive cues	AI-generated fake reviews	6.86		
	Human-generated fake reviews	10.52	3.66	0.00***
	Authentic reviews	10.57	3.71	0.00***

^a Note: ***Significant at .001 level; ** Significant at .01 level; * Significant at .05 level

compared to both human-generated fake reviews and authentic reviews, challenging conventional assumptions. Existing research has long posited that lower readability serves as a key discriminator to distinguish fake from authentic reviews (Moon et al., 2021). However, our

Table 6
Independent sample t-test results.

Variable	Review group	Mean	SD	t	p
Mechanicalness	AI-generated fake reviews	3.39	1.22	73.16	0.00***
	Authentic reviews	3.19	0.714		
Empathy	AI-generated fake reviews	0.36	0.17	-39.55	0.00***
	Authentic reviews	0.39	0.16		

Note: ***Significant at .001 level; ** Significant at 0.01 level; * Significant at .05 level.

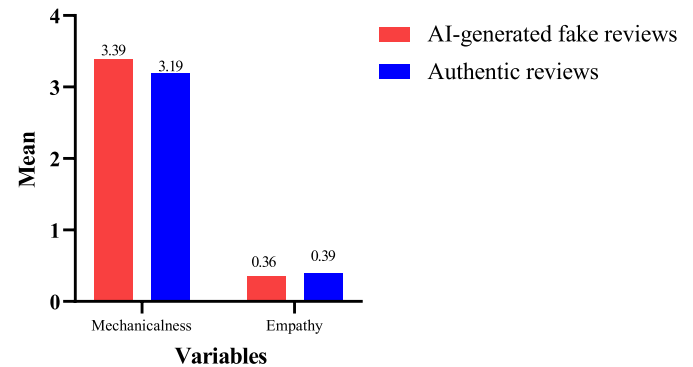


Fig. 3. Linguistic differences between AI-generated fake reviews and authentic reviews.

findings emphasize the need for caution when interpreting readability metrics currently. Specifically, it suggests that high-readability online reviews require additional scrutiny to determine if they are AI-generated fake reviews. On the other hand, this study validated the utility of specificity as a discriminative metric (Banerjee and Chua, 2017a). Findings showed that AI-generated fake reviews demonstrated lower specificity compared to both human-generated fake reviews and authentic reviews. This indicates that specificity remains a reliable indicator for distinguishing authentic reviews from fake reviews.

Secondly, this study uncovers a dual counterintuitive pattern in AI-generated fake reviews: they exhibit significantly lower levels of exaggeration and negligence compared to human-generated fake reviews and authentic reviews. Typically, fake reviews are more exaggerated than authentic ones (Wang et al., 2022). This finding extends prior research by suggesting that unusually subdued or neutral-toned reviews may also signal AI generation. Similarly, AI-generated fake reviews exhibited lower negligence compared to both human-generated fake reviews and authentic reviews. Prior research posits that fake reviews are more negligent due to psychological factors (Banerjee and Chua, 2017a). However, AI's computational rigor and rule-based structure may disrupt this pattern, resulting in reduced negligence (Jackson and Panteli, 2023). These findings highlight the need for heightened scrutiny when encountering reviews with both low exaggeration and low negligence, as they may indicate AI-generated fake reviews.

Finally, AI-generated fake reviews were found to be more mechanical but have less empathy compared to authentic reviews. This phenomenon can be attributed to AI's reliance on pre-trained algorithms and large-scale language data, resulting in the production of content that is organized and coherent (Liu et al., 2023). Although the output of AI-generated tools may vary across different topics, their underlying structural integrity and logical consistency remain remarkably stable. Especially in situations such as online reviews, where there is a dynamic communication context (Kim et al., 2023b), AI still provides well-structured and logical responses during the generation process. This implies that, through the examination of the structural and logical characteristics of AI-generated fake reviews, it becomes more feasible to

differentiate them from genuine reviews. Nevertheless, AI generation falls short of human beings in terms of feeling cues as predicted (Chen et al., 2022). It is still difficult for AI-generated tools to demonstrate human-like perception and empathy in textual content. We supposed that they may be more inclined to provide standard, templated responses than content that conveys rich sensory and empathic knowledge.

8. Conclusions

Investigating the linguistic characteristics of AI-generated fake reviews carries important implications for maintaining healthy business ecosystems and protecting consumers' benefits. The emergence of AI-generated fake reviews challenges existing detection rules and methods. Through comparative analysis among AI-generated fake reviews, human-generated fake reviews and authentic reviews, we identify that AI-generated fake reviews paradoxically demonstrate higher comprehensibility yet lower specificity, reduced exaggeration, and diminished negligence. Furthermore, building on fundamental differences between human and AI writing processes, our analysis reveals these AI-generated fake reviews exhibit greater mechanicalness and notably weaker empathic engagement compared to authentic reviews. These findings not only reshape our understanding of fake review indicators in the AI era but also provide actionable insights for developing more robust detection methods. By establishing these distinctive linguistic cues, the study offers practical guidance for identifying AI-generated fake reviews, ultimately helping mitigate its potential harm to online marketplaces and consumer trust.

8.1. Theoretical contributions

This study provides the following two key theoretical contributions. First, this research extends the literature on fake reviews by investigating the linguistic distinctions between AI-generated fake reviews and human-generated fake reviews. Prior studies have primarily focused on differentiating AI-generated fake reviews from authentic reviews, with limited attention paid to the contrast between AI-generated and human-generated fake reviews. Addressing this gap, our study systematically examines the differences between the two types of fake reviews, revealing the inadequacy of existing fake review detection rules in identifying AI-generated fake reviews. These findings provide a theoretical foundation for optimizing fake review detection methods and technologies.

Second, this study innovatively integrates the theoretical model of textual differences in online reviews with AI capabilities. While prior research has identified linguistic cues for distinguishing AI-generated fake reviews from authentic reviews, these studies lack a systematic comparison of linguistic variations and fail to account for the unique characteristics of AI. By applying the theoretical model to the analysis of AI-generated fake reviews, this work expands its applicability and theoretical depth. Furthermore, by incorporating AI capabilities, we explore linguistic features that may enhance the detection of AI-generated fake reviews, offering a novel perspective for identification. These insights advance research on AIGC and open new avenues for future AIGC studies.

8.2. Practical implications

The findings of this study carry significant practical implications for policymakers, platforms, retailers, and other specific party seeking to mitigate the risks posed by AI-generated fake reviews.

First, for policymakers and regulatory authorities, the study's findings underscore the necessity to develop targeted regulatory frameworks and clear penalty measures to address AI-generated fake reviews. Regarding regulatory frameworks, it is necessary for policymakers to design additional detection rules based on proposed indicators such as

readability and exaggeration as AI-generated fake reviews deviate from the traditional detection rules for human-generated fake reviews. This involves establishing disclosure norms for AI-generated content and online reviews, as well as refining criteria for identifying AI-generated fake reviews. Regarding penalty measures, policymakers can design punitive measures tailored to different entities. For example, transparent and standardized penalty procedures should be established for retailers and platforms that fail to adopt optimized detection methods. Entities involved in disseminating or generating fake reviews should face tiered sanctions based on the severity of their violations.

Second, for platforms and their retailers, the findings of this study provide a solid foundation for optimizing fake review detection systems. Integrating language recognition cues specific to AI-generated fake reviews is the key to accurately identifying them. Based on our research results, platforms and retailers can develop a two-tiered detection model. The first tier can build on existing detection methods by leveraging a text-differentiation analysis framework. Guided by the study's findings, it screens and flags reviews for further examination. For instance, the results indicate that AI-generated fake reviews do not exhibit the same high level of negligence as human-generated fake reviews, a characteristic that is closer to authentic reviews. Thus, reviews with low negligence should be selected for additional scrutiny. The second tier develops specialized identification methods tailored to the characteristics of generative AI, namely high mechanicalness and low empathy. By quantifying and comparing the mechanicalness and empathy metrics of online reviews, this tier offers a practical approach to enhance the identification of AI-generated fake reviews.

Third, for customers, the linguistic cues proposed in this study can help them identify AI-generated fake reviews. Since the findings of this research are all grounded in text analysis, users can perform simple vetting of reviews by leveraging our findings. For example, the specificity of a review, whether it mentions real-world usage scenarios or personal experiences, can serve as a key indicator. By examining these elements, potential consumers can distinguish AI-generated fake reviews from authentic reviews. More importantly, it is feasible for users to make careful and prudent judgments about fake reviews. For instance, they can use the analytical tools mentioned in this study to quantify reviews and make analyses and judgments based on the study's findings. This approach not only bridges the gap between academic research and practical application but also equips consumers with tangible tools to navigate the evolving landscape of AI-generated fake reviews.

8.3. Limitations and future research

This study has some limitations that may inform future studies. First, this study focuses on the linguistic cues of AI-generated fake reviews, as language serves as the most direct carrier for information transmission in online reviews. This focus may limit the integration of non-linguistic features for fake review detection. With the increasing quantity and scaling trend of AI-generated fake reviews, future research can delve into non-linguistic features, such as review posting times, account profiles, and other information. Future research could incorporate non-linguistic features to both reinforce the findings of this study and enable cross-validation, thereby establishing a more comprehensive framework for identifying AI-generated fake reviews.

Second, this research is conducted on the online reviews in English, which may limit its applicability in other linguistic environments. Although this study provides referable indicators and quantitative methods, the research results may not fully cover the characteristics of non-English reviews. Future research can expand the scope to online reviews in different languages by constructing multilingual datasets. On this basis, future studies can replicate the design of this study to validate existing findings and develop tailored methods for identifying fake reviews in different languages, assisting platforms and businesses in effectively detecting AI-generated fake reviews in diverse linguistic scenarios.

CRediT authorship contribution statement

Yuxin Zhao: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Siyi Tang:** Investigation, Data curation. **Hongyu Zhang:** Writing – review & editing, Supervision, Funding acquisition. **Long Lyu:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that there is no conflict of interests regarding the publication of this paper.

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Data availability

Data will be made available on request.

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